

JBM9204 INTRODUCTION TO BIOMETRIC SYSTEMS

UNIT-II

FINGERPRINT IDENTIFICATION TECHNOLOGY

Biometrics using Fingerprint

Fingerprint biometrics is the systematic study and application of unique physical attributes inherent in an individual's fingerprints. Representing a more dependable identification method than traditional passwords or identity cards, fingerprint biometrics eliminates the issues of misplacement, forgetfulness, or theft. The distinctive nature of each person's fingerprint ensures a robust barrier against unauthorized access to secure data.

General Description of Fingerprints

What is a Fingerprint?

Fingerprints are the unique pattern of ridges, or dermatoglyphs, that appear on the skin of the hands and feet. They can have an arching, looping, or swirling pattern, but the exact patterns differ from person to person.



Why do we have fingerprints?

The ridges and grooves on our fingertips are more than just unique patterns, they provide essential functions for daily life.

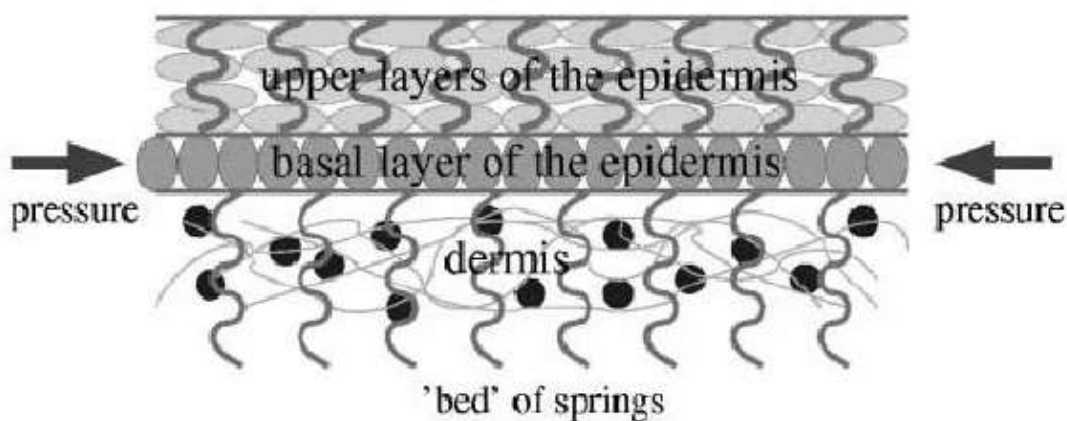
1. **Improved Grip** - One key reason for fingerprints is to improve grip. The raised ridges increase friction between our skin and objects, helping us hold things securely.

2. **Enhanced Touch Sensitivity**- Fingerprints also enhance our sense of touch. The ridges amplify vibrations when our fingers interact with surfaces, sending more signals to the brain.

3. **Protection and Adaptation**- From an evolutionary standpoint, fingerprints likely developed to protect our fingertips. The ridges help channel away water or sweat, maintaining a secure grip even in damp conditions. This feature would have been essential for early humans climbing trees, handling tools, or gathering food.

Development of Fingerprints in womb

Fingerprints start forming around **10–15 weeks of gestation** in the fetus. The basal layer of the epidermis grows faster than the layers above and below it. This causes it to buckle and fold, creating the unique ridge patterns (arches, loops, and whorls). As the fingers grow, new ridges and branches develop. Between 17 and 19 weeks gestation, a layer of keratin coats the surface of the skin. Smaller, secondary ridges form out of uneven keratin growth.



The formation is influenced by genetic factors (DNA), but also by random physical forces in the womb—like the fetus's position, the density of amniotic fluid, blood pressure, oxygen levels in the blood, nutrition of the mother, hormone levels and movements. The pattern of fingerprints established by **19 weeks** remains consistent as the child grows.

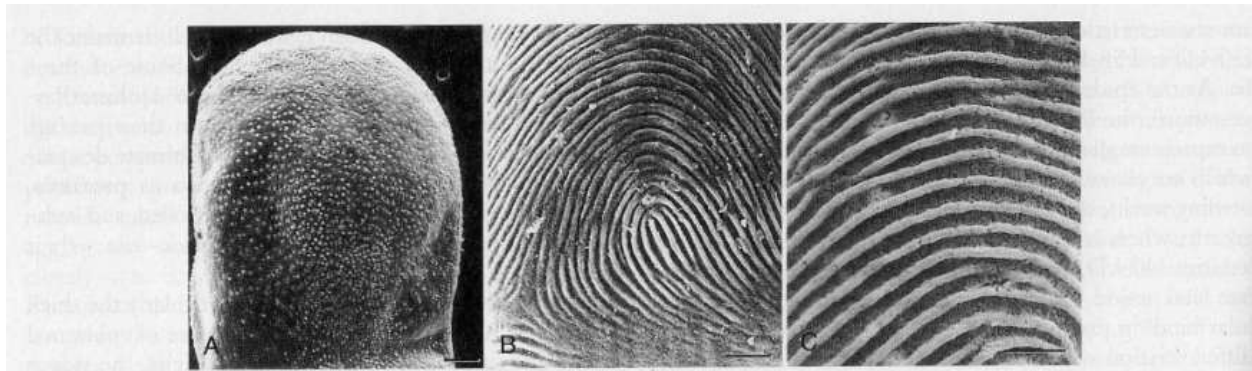


FIGURE 9-6. Scanning electron micrographs of human digital palmar skin in a 14-week fetus. **A**, Low-power view of the palmar surface of a digit. **B**, Epidermal surface of the dermis of the fingertip showing the primary dermal ridges. **C**, Basal surface of the epidermis showing the epidermal ridges. Bars, 100 μm . (From Misumi Y, Akiyoshi T:

Why fingerprints are unique?

Fingerprint pattern formation consists of two components: developmental and genetic.

The combination of **genetic blueprint + random physical conditions** creates a one-of-a-kind pattern for every individual. Even identical twins (with the same DNA) have different fingerprints because the tiny, random environmental factors in the womb—like pressure, growth stresses, and exact timing of skin development—cannot be replicated.

Aspects of fingerprints that are analyzed include:

- **Patterns:** The general pattern or type of fingerprint (arch, loop, or whorl) is inherited through genetics.
- **Minutiae:** The precise details of the ridges, known as minutiae, are influenced by random and unpredictable factors such as pressure, blood flow, and position in the womb during development.
- **Ridges:** Each ridge in a fingerprint contains several minutiae points, which can be bifurcations (where one ridge splits into two) or ridge endings.

Principles of Fingerprints

Fingerprints follow three basic principles:

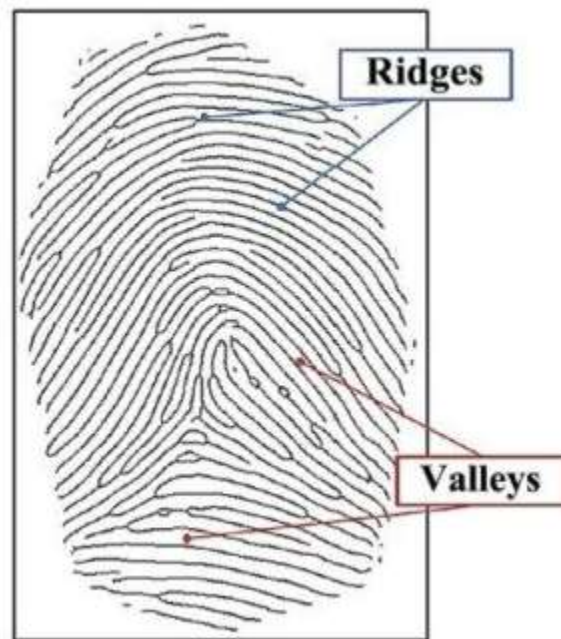
1. **PATTERN TYPE:** All fingerprints can be classified into pattern types.
2. **UNIQUENESS:** Each fingerprint is unique, no two people have the same fingerprints. Each person's fingerprints differ from finger to finger.

3. **PERMANENCE:** Fingerprints are unchanging from 3 months fetal life until decomposition. Scars are the only exemption to this rule.

Structure of the Fingerprint

A fingerprint is made up of friction ridges, the raised skin patterns on the fingertips, palms, and soles.

A visual inspection of friction skin reveals a series of lines corresponding to hills (ridges) and valleys (grooves).



Ridges:

- These are the raised, dark lines on the skin of the fingertip.
- They are formed by the upper layer of the epidermis and help with grip and touch sensitivity.
- Ridges create the unique fingerprint patterns—**arches, loops, and whorls**—used for identification.
- Each ridge has **sweat pores** that leave impressions (latent fingerprints) on surfaces.

Valleys:

- These are the lighter, lower spaces between the ridges.

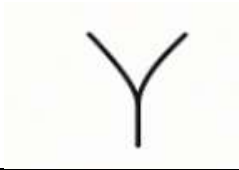
- Valleys separate the ridges and give fingerprints their distinctive patterns.
- They play a role in channeling sweat away from the skin surface, maintaining grip.

Ridge Characteristics (Minutiae Points)

Ridge characteristics, also known as **minutiae**, are the fine details in the patterns of friction ridges found on the skin's surface, primarily on the fingers, palms, and soles. These unique ridge features are crucial in forensic fingerprint identification. No two fingerprints share the same arrangement of minutiae, making them a highly reliable method for personal identification. Forensic experts and automated systems like **AFIS (Automated Fingerprint Identification System)** use these minutiae to match fingerprints for crime scene investigation, personal identification, and legal proceedings.

- Fingerprint recognition, whether done manually by a human expert or automatically by a machine, is predominantly feature-based (as opposed to image-based) and the features used have a physical interpretation.
- The terms ***feature-based*** and ***image based*** are widely used in the computer vision literature to indicate the methods used for representing and matching images such as fingerprints.
- A feature-based method, extracts explicit features from the image and encodes these features into a feature set, which is subsequently used for matching.
- An image-based method, directly uses the image for matching without explicitly extracting any features from it.



Minutiae Type	Structure	Description
Crossover		A point where two ridges cross each other.
Core		The center or focal point of the fingerprint pattern, usually in a loop or whorl (Innermost loop of the fingerprint).
Bifurcation		A point where a single ridge splits into two ridges.
Ridge Ending		The point where a ridge abruptly stops.

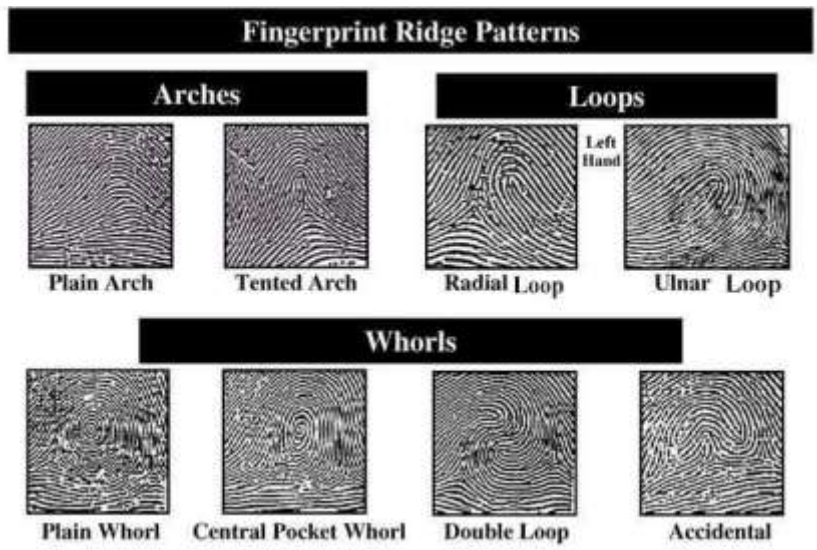
Island		A small, isolated ridge or a very short ridge segment. (Very short ridge that stands alone)
Delta		A triangular region where ridge patterns diverge; usually found in loops and whorls.
Pore		Tiny openings on the ridges where sweat is released; visible in high-resolution fingerprint images.
Hook (Spur)		A small ridge extending from a longer ridge but not branching off completely.
Trifurcation		A ridge that splits into three distinct ridges, forming a "trident" or fork-like shape.
Opposed Bifurcation		Two bifurcations facing each other where ridges split in opposite directions.
Incipient Ridge		A faint, underdeveloped ridge, shorter or thinner than surrounding ridges.

Ridge characteristics, or minutiae, form the foundation of fingerprint analysis in forensic science. These unique details—such as ridge endings, bifurcations, lakes, and cores—are critical to personal identification. The arrangement of these minutiae within a fingerprint is what makes it unique to each individual.

Fingerprint Ridge Patterns

There are three main types of fingerprint patterns with several subgroups of patterns:

Arch
Loop
Whorls



1. **Arch patterns** account for approximately 5-15 percent of fingerprint patterns. Arch patterns can be further categorized into two subgroups:

a. **Plain arches**

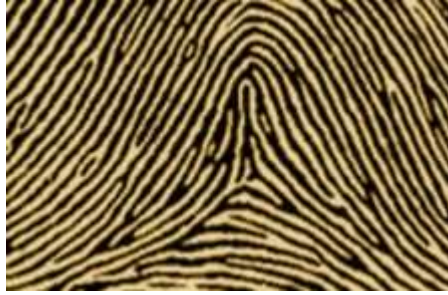


It consists of ridges that flow from one side of the finger to the other in a smooth, wave-like manner. This pattern lacks the complexities of loops or whorls and has no deltas (triangular formations created by ridge paths).

Plain arches are the least common fingerprint type, appearing in only about 5% of the population. Their simplicity makes them easy to identify.

The ridges rise slightly at the center but form no sharp angles or curves.

b. Tented arches



In this pattern, the ridges rise sharply at the center, creating a peak or tent-like structure. This peak gives the pattern its name.

Unlike plain arches, tented arches have a distinct central ridge that forms an angle or spike. This makes them more complex and slightly more common than plain arches, but still relatively rare.

Tented arches account for about 1-2% of all fingerprints.

The tented arch's unique structure makes it a fascinating pattern in fingerprint analysis. Its sharp rise sets it apart from other types, making it easily distinguishable.

2. **Loop patterns** account for 60-65 percent of fingerprint patterns. The loop pattern is the most common. There are two subgroups (automated systems simply use the terms right slant or left slant):

a. Ulnar (ridges flow toward the little finger)



Ulnar loops are the most common fingerprint pattern, present in about 60% of individuals. These loops curve inward toward the little finger, following the direction of the ulna bone in the forearm.

The ridges in an ulnar loop enter and exit on the same side of the finger, forming a loop. Ulnar loops are easily recognized and widely studied due to their high occurrence rate.

They serve as a foundational pattern in the classification and identification of fingerprints.

b. Radial (ridges flow toward the thumb)



Radial loops are fingerprint patterns that curve toward the thumb. They are named after the radius bone, located on the thumb side of the forearm.

The ridges in this pattern enter and exit on the same side of the finger, forming a loop that flows in the direction of the thumb. Radial loops are less common than ulnar loops and are found in a smaller percentage of the population.

They are unique due to their specific directional flow, which distinguishes them from other loop patterns. In fingerprint classification, the direction of the loop is critical for accurate identification.

Radial loops highlight the importance of understanding ridge flow when analyzing prints.

3. Whorl patterns account for 30-35 percent of fingerprint patterns. Whorls can further be categorized into four subgroups:

a. Plain



Whorls are circular or spiral ridge patterns that form complete loops. They are one of the most easily recognizable fingerprint types. Whorls appear in about 30% of the population, making them a relatively common pattern.

The defining feature of a whorl is its central ridge, which completes at least one full circle. Whorls often include two deltas, located on opposite sides of the circular ridge pattern. Their clear, circular design makes them easy to classify and compare.

b. Double loop—sometimes called lateral loops



Double loops are intricate patterns that feature two separate loops intertwining around each other.

This unique design makes double loops stand out as one of the more complex fingerprint types. The ridges in a double loop form two distinct centers, creating a pattern that resembles an 'S' or two opposing loops.

This type appears in about 10% of fingerprints, making it less common than ulnar loops but more intricate. Double loops are often a subject of interest in forensic science due to their complexity.

c. Central pocket loop



Central pocket loops combine elements of loops and whorls. The pattern features a loop with a circular or spiral whorl at the center.

This pattern is less common than basic loops but more intricate.

The ridges in a central pocket loop form a distinctive circular shape within the loop, adding complexity to the overall design.

d. Accidental



Accidental fingerprints are the rarest and most complex type.

They combine features of two or more patterns, often including elements of loops, whorls, and arches. The irregularity of accidental fingerprints makes them challenging to classify.

They don't fit neatly into any single category, which is why they are labeled as 'accidental'.

Their unique structure makes them highly distinctive, providing a reliable basis for individual identification.

FINGERPRINT RECOGNITION SYSTEM

Fingerprint recognition is a biometrics-based pattern recognition system widely used to identify or authenticate a person by determining the authenticity of the fingerprint, an impression or mark made on a surface by one finger.

A fingerprint recognition system (FRS) works because of the uniqueness of the fingerprints since no two people have identical fingerprint patterns. They are unique to everyone, even to identical twins, and do not change over time.

Because a fingerprint is an individual characteristic, remains unchanged during an individual's lifetime, and has general ridge patterns that can be classified systematically based on pattern types, size, and position of the finger, it is a widely used biometric characteristic.

A fingerprint-based biometric system is called either a verification system or an identification system, depending on the context. A verification system authenticates one's identity by comparing the fingerprints to a pre-stored biometric template(s). It conducts a one-to-one comparison to determine whether the identity claimed by the individual is true. Meanwhile, an identification system recognizes an individual by searching the entire template database for a match. It conducts one-to-many comparisons to establish the identity of the individual.

With the widespread use of biometric scanners on smartphones and a growing number of security breaches, fingerprint recognition is increasingly replacing traditional authentication methods such as passwords and PINs. Fingerprint recognition offers an alternative solution to the task of personal authentication or identification since unlike passwords or biometric traits cannot be forged, lost, or forgotten.

A typical biometric system consists of four modules:

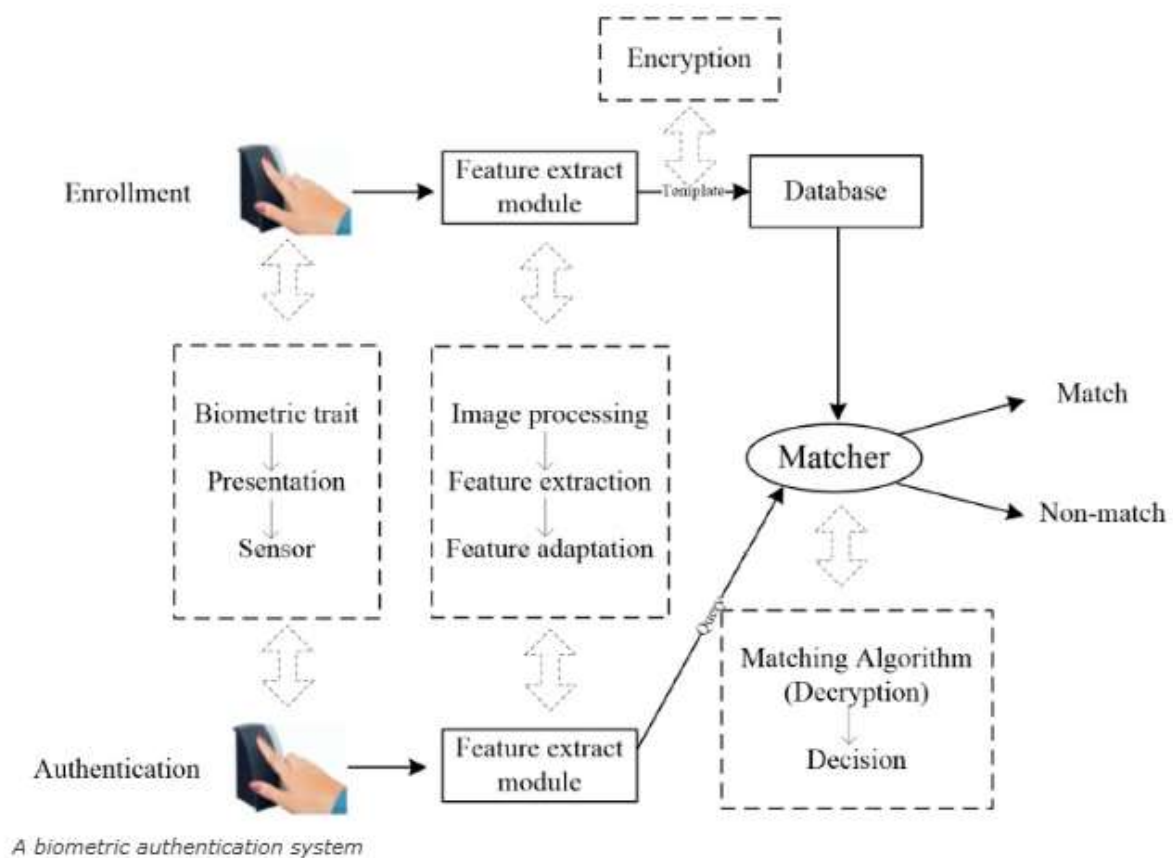
A sensor module, a feature extraction module, a template database, and a matching module. The sensor module acquires the biometric image, from which a set of global or local features are then extracted by the feature extraction module. These features are stored in the template database as template data. Finally, a

matching module compares the query and template data to arrive at a match or non-match verdict.

A typical biometric system carries out authentication in two stages:

Enrollment and verification.

In enrollment, a user presents the fingerprint image to be acquired by the sensor module. Certain features are extracted, and further adapted or transformed to generate template data for comparison in the next verification stage. In verification, the fingerprint image of a query is collected by the sensor module and then compared with the existing template data.



Similarly, a fingerprint verification system has four main modules such as;

- Fingerprint sensing that acquires the fingerprint of an individual through a fingerprint scanner, producing a raw digital representation. Three families of electronic fingerprint sensors (such as solid-state or silicon sensors, optical, and ultrasound) are used for the acquisition of fingerprint images.

- Preprocessing that simplifies the input fingerprint for feature extraction.
- Feature extraction, where the fingerprint is processed to generate discriminative properties, called feature vectors.
- Matching, in which the feature vector is compared against existing templates stored in a database.

There are currently **various types of fingerprint matching techniques** that are responsible for the recognition of fingerprint patterns. A better matching technique provides a better result during matching. Some of the fingerprint matching techniques and identification keys are:

Minutiae based

Pattern based

Feature based

ROI (Region of Interest) based

Correlation based

Statistics based

ONNC (Optical Neural Network Computer)

The main parameters characterizing a digital fingerprint are

Resolution: It indicates the number of dots or pixels per inch (dpi). Though 500 dpi is the minimum resolution for FBI-compliant scanners, 250 to 300 dpi is the minimum resolution needed for extraction algorithms to locate the minutiae in fingerprint patterns.

Area: It is the size of the rectangular area sensed by a fingerprint scanner. The bigger the area, the more ridges, and valleys are captured, and the more the fingerprint becomes distinctive.

Dynamic range (or depth): It denotes the number of bits used to encode each pixel's intensity value. The FBI standard for depth of the pixel bit is 8 bits, yielding 256 gray levels.

Geometric accuracy: It is specified by the maximum geometric distortion introduced by the acquisition device, and expressed as a percentage concerning x and y directions.

Image quality: It is not easy to precisely define the quality of a fingerprint image. Image quality is key to identify the intrinsic finger status. If the fingers are too moist or too dry, or when they are incorrectly presented, most of the scanners produce poor quality images.

Benefits of FRS

- High uniqueness, accuracy, performance, and stability.
- Easy to use and reliable than other techniques.
- Good performance, low cost, and small size.
- Have approximately 50% popularity in the market as compared to other techniques.
- Has an irreplaceable role of latent prints in crime investigation.
- Enable government systems to use large databases of fingerprints with a criminal history.

Key challenges in FRS

- Mismatching of fingerprints due to physical distortions, finger injuries and cuts.
- Vanishing or fading fingerprint due to jobs like cleaning.
- Mismatching due to displacement or rotation while scanning the finger.
- Unauthorized access due to finger plasticity or clay printing.
- Variability between impressions due to skin conditions, noise in the sensor.
- Small area fingerprint sensors, providing less information.
- Fake fingerprints i.e., clays or dummy printing.

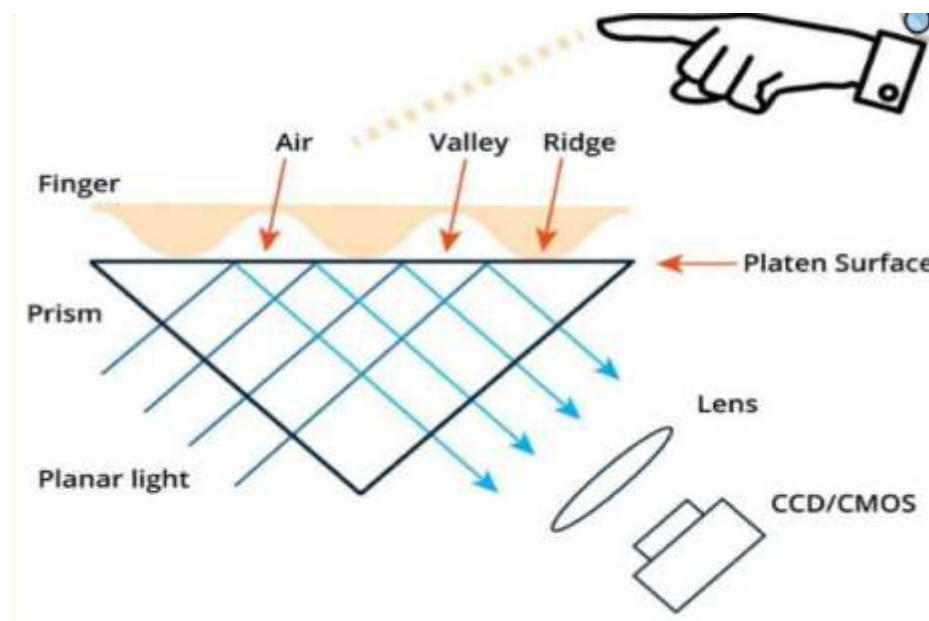
FINGERPRINT SENSORS USING RF IMAGING TECHNIQUES

Fingerprint sensing is one of the most widely deployed techniques in biometric sensing. Fingerprint sensors can be categorized by the way the user interacts with them, such as contact, contactless, slap, partial, etc. Several sensing mechanisms have been used to detect the finger's ridge-valley structure. Based on the underlying technology the sensors utilize, they are categorized into

- (i) Optical sensors;
- (ii) Capacitive sensors
- (iii) Ultrasonic sensors;
- (iv) Thermal sensors; and
- (v) Pressure sensors.

OPTICAL SENSING

This is the oldest 'live-scan' fingerprint sensor, in which a glass prism is illuminated when a finger touches the prism.



The above figure shows the principle behind a typical optical sensor where ridges absorb the light, while the valleys allow the light to be reflected, and the reflected light is caught by CCD or CMOS sensors. Optical sensors are **low-cost fingerprint sensing devices comprised of the sub-techniques of reflection, transmission, sweep, TFT and electro-optical processes**. The primary technique the sensor uses is **frustrated total internal reflection (FTIR)** with an arrangement of a glass prism, a laser light and a CMOS or CCD sensor. It is not easy to arrange all of them in a portable form due to the very large focal length of small lenses. Further, it is not very easy to fool FTIR, and therefore, the scanners are vulnerable to imprecise fingerprint imaging.

Advantages

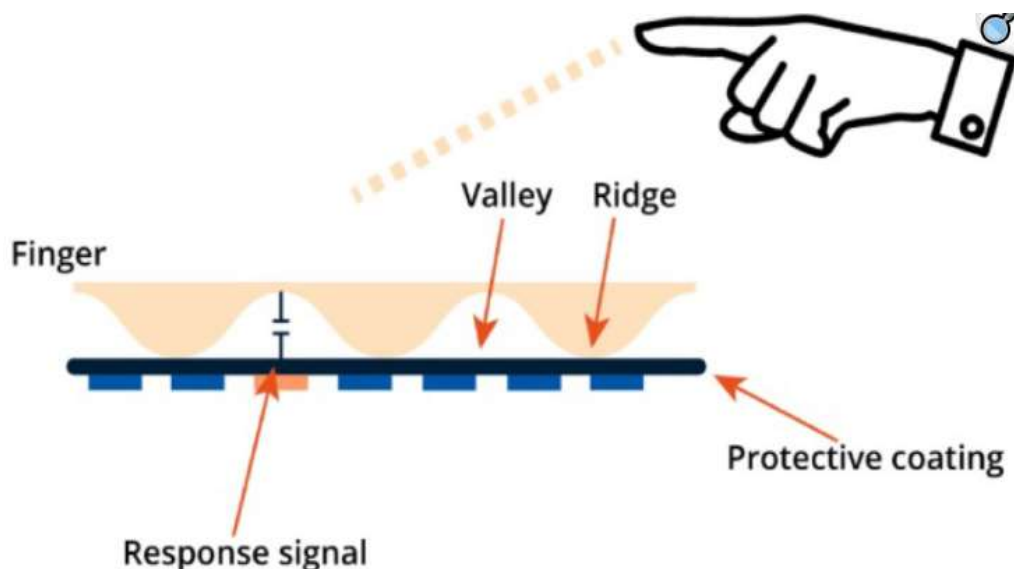
- Low cost,
- Produces good-quality images within a large sensing area and
- preventing electro-static discharge (ESD)

Disadvantages

- It cannot be shortened further due to the distance between prism and sensor, which limits the miniaturization.
- sensitive to the contamination on fingers or platen, finger dryness and moisture

CAPACITIVE SENSING

To overcome the drawbacks existing in optical sensing, capacitive silicon sensing has emerged. Capacitance is an electrical property present between two conductive surfaces. In capacitive sensing, many capacitance plates are embedded in a chip. When a finger is placed, an electric charge is generated between the plate and finger. The sensor captures the capacitance varieties from the ridge-valley pattern.



Advantages

- Smaller in size,

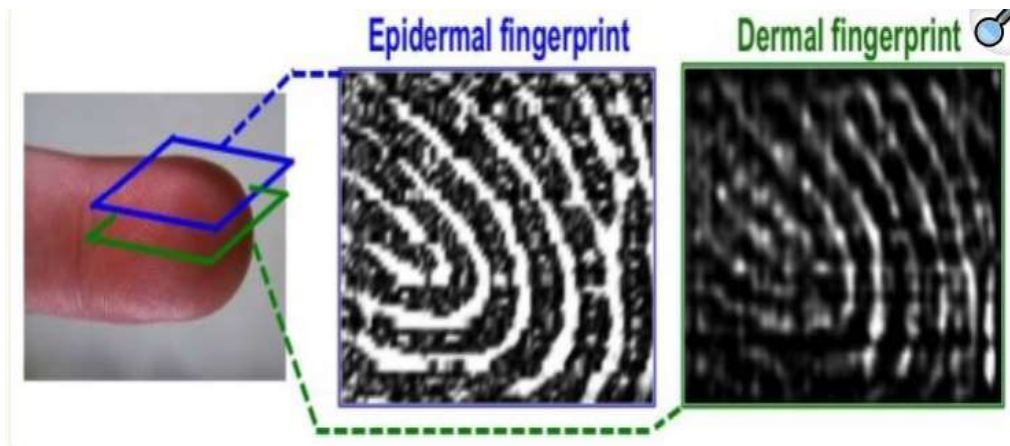
- Consumes less power and
- User friendly.

Disadvantages

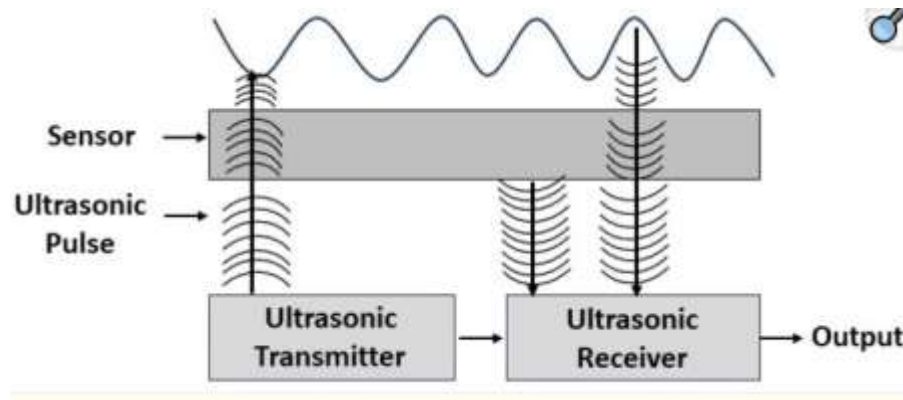
- Vulnerable to strong external electrical fields,
- ESD and
- Expensive

ULTRASONIC SENSING

Ultrasonic fingerprint sensing can be an alternate means for the existing fingerprint sensing modes. The principle behind the ultrasonic sensing is a medical ultrasonography where high frequency sound waves penetrate into the skin's epidermal layer. Since dermal and epidermal layers have similar features, as illustrated in the Fig, the reflected measures are used to capture the finger image using piezoelectric materials.



The sensor has sender and receiver modules where the former sends acoustic signal towards the finger, while the latter acquires the results when these signals backlash the fingerprint surface.



Piezoelectricity is the fundamental of ultrasonic sensing, and later on, lead zirconate titanate (PZT), which is strong and has piezoelectric properties, was discovered.

The overall ultrasonic sensing technique is divided into two categories:

- (i) Ultrasound imaging techniques where pulse-echo imaging and impediography methods are approached and
- (ii) Transducer technologies where piezocomposite transducers, capacitive micromachined ultrasonic transducers (CMUT) and piezoelectric micromachined ultrasonic transducers (PMUT) are explored.

The **advantage** of using ultrasonic sensing is that it has higher reliability compared to other sensing schemes as it computes the difference between the acoustic impedance of the finger ridge-valley structure. However, a shortcoming of higher cost was experienced.

The **significance of the ultrasound sensing** over other technologies is that it has the capability to acquire a large volume of the finger, supports wet fingers and results in several benefits:

- (i) Distinctive feature extraction;
- (ii) Flexibility of finger touch location due to the large area;
- (iii) Convenience of use;
- (iv) Lack of vulnerability from surface contamination and humidity.
- (v) pulse-echo ultrasonic imaging measures images at multiple depths from the sensor and beneath the epidermis, which resulted in the image's resistance to spoof attacks.

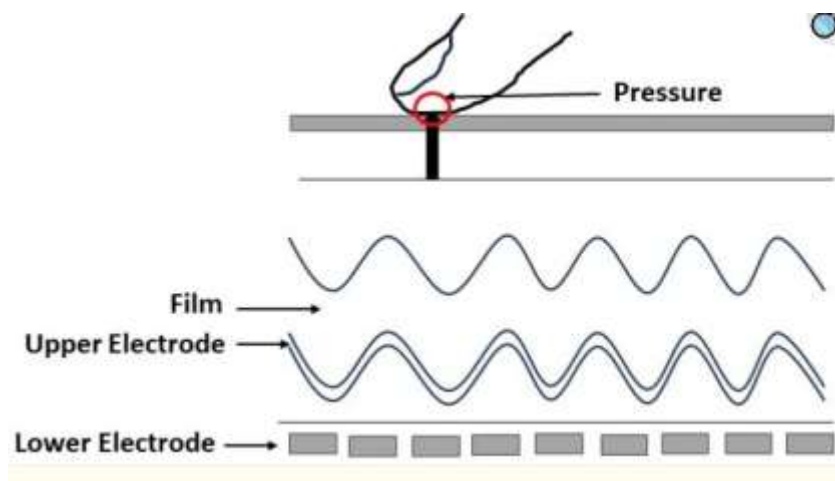
TEMPERATURE DIFFERENTIAL SENSING

This sensor works by detecting temperature differences that happen when two surfaces touch each other.

It contains a special material called a pyroelectric material, which has a unique property:

When the temperature changes, it produces a small electric current (or voltage).

So, when the sensor touches something (like a finger), the temperature difference causes the material to create a voltage signal — and this signal is what the sensor uses to detect the contact.



Finger print Sensors using RF Imaging

Fingerprint sensors using **RF (Radio Frequency) imaging techniques** rely on electromagnetic waves instead of just optical or capacitive methods.

RF fingerprint sensors use radio frequency signals to penetrate the outer layer of skin (epidermis) and capture the fingerprint pattern from inner layers (dermis). This makes them different from traditional optical or capacitive sensors, which only read the surface pattern.

Main Components of RF Fingerprint Sensor

1. RF Emitter (Transmitter)

- Generates low-power radio frequency waves (typically in the range of MHz to GHz).

- Sends the RF signal toward the finger placed on the sensor surface.

2. RF Receiver (Antenna or Sensor Array)

- Detects the reflected RF signals that bounce back from different layers of the skin.
- Measures differences in reflection caused by ridges and valleys.

3. Antenna Array

- A grid of micro-antennas is used for transmitting and receiving RF signals.
- Helps in capturing a detailed 2D/3D map of the subsurface structure.

4. Dielectric Layer

- Protective layer between the finger and the antenna.
- Designed to allow RF waves to pass through but protect electronics.

5. Signal Processing Unit

- Converts the RF reflection data into a digital image.
- Uses algorithms to extract minutiae points (ridge endings, bifurcations).

6. Microcontroller / Processing Chip

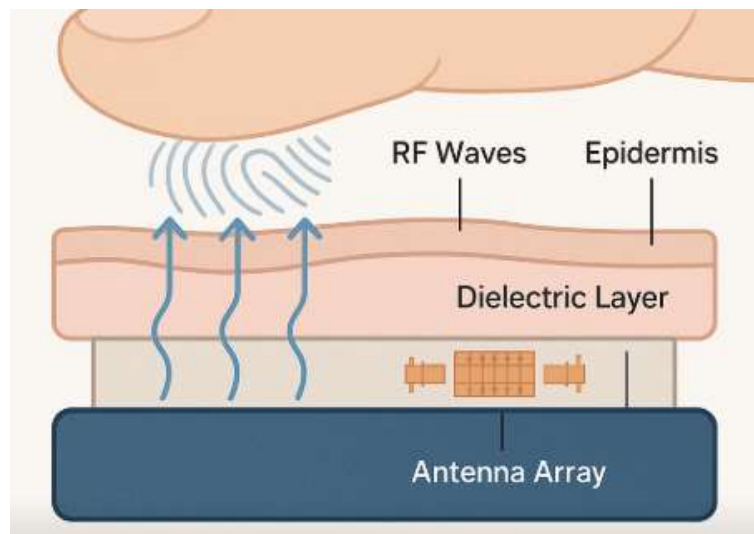
- Handles the data analysis, encryption, and communication with the main device (phone, security system).

7. Power Management Unit

- Regulates power for RF transmission and signal processing.
- Important because RF imaging needs slightly higher energy than capacitive sensors.

8. Interface Module

- Connects the sensor with the host system (e.g., smartphone, access control).
- Often supports SPI, I²C, or proprietary protocols.



Advantages Over Optical/Capacitive

- Works with Dirty, Wet, or Damaged Fingers (since it scans deeper layers).
- Higher Security: Cannot be easily fooled by surface replicas or fake fingerprints.
- Better Accuracy: Reads subsurface ridge structure.

Applications

- High-security devices (e.g., military, government).
- Some advanced smartphones (though rare due to cost).
- Access control in harsh environments.

FINGER PRINT FEATURE PROCESSING TECHNIQUES

Finger print and Finger print Technology

Fingerprints are unique patterns of ridges, whorls, and loops found on the fingertips of humans. These patterns, formed during fetal development and unchanged throughout life, serve as distinctive identifiers for individuals. Fingerprint technology utilizes these unique patterns to authenticate and verify identities in various applications. By capturing and analyzing the unique characteristics of an individual's fingerprint, this technology enables secure access to devices, buildings, and sensitive information. Fingerprint recognition systems employ sophisticated

algorithms and sensors to scan and match fingerprints, offering a reliable and convenient method of biometric authentication.

Benefits of Fingerprint Technology

Fingerprint technology offers a multitude of benefits, making it a preferred choice for biometric authentication.

1. Its unparalleled accuracy ensures reliable identity verification, minimizing the risk of unauthorized access.
2. Fingerprint authentication provides convenience for users, eliminating the need to remember complex passwords or carry physical tokens. Additionally, the unique nature of fingerprint patterns enhances security, as each individual possesses distinct biometric characteristics.
3. The speed of fingerprint authentication enables quick access to secured systems or facilities, enhancing operational efficiency.
4. Fingerprint technology is versatile, with applications ranging from personal devices to corporate security systems, making it adaptable to diverse environments and use cases.
5. The combination of accuracy, convenience, security, speed, and versatility makes fingerprint technology an invaluable tool for ensuring secure access to digital services and protecting sensitive information.

Working of Fingerprint Technology

Fingerprint technology, a subset of biometric authentication, relies on the unique patterns and ridges found on an individual's fingers to verify their identity.

The process begins with **capturing an image** of the fingerprint using specialized scanners, which **extract specific features** such as ridge endings and bifurcations. These features are then used to create a **digital template** representing the fingerprint biometric, forming a mathematical model of its unique attributes. When a user attempts authentication, a new fingerprint image is **captured and compared** to stored biometric templates using pattern-matching algorithms. The system analyzes **similarities and differences** between the templates to determine authentication, granting access if a close match is found, and denying it otherwise. This method offers high accuracy and security in various applications, including security systems and mobile devices.

Fingerprint Recognition Steps

The process involves several key steps:

1. **Image Capture:** The first step is to capture an image of the fingerprint. This is typically done using specialized fingerprint scanners, which may utilize different technologies such as optical, capacitive, or ultrasound.
2. **Feature Extraction:** Once the fingerprint image is captured, the system extracts specific features from it. These features include ridge endings, bifurcations, and other unique characteristics of the fingerprint.
3. **Template Creation:** The extracted features are then used to create a digital template or representation of the fingerprint. This template is essentially a mathematical model that captures the unique attributes of the fingerprint.
4. **Matching Process:** When a user attempts to authenticate their identity using their fingerprint, the system captures a new image of the fingerprint. This image is processed to create a new template, which is then compared to the stored templates in the system's database.
5. **Pattern Matching:** The system employs pattern-matching algorithms to compare the new template with the stored templates. It looks for similarities and differences between the two templates to determine whether they match.
6. **Decision:** Based on the results of the pattern-matching process, the system decides the authenticity of the fingerprint. If the new template closely matches any of the stored templates, the user is granted access. Otherwise, access is denied.

1. Fingerprint Acquisition (capture)

- The first step in fingerprint recognition is **image acquisition** - the process of capturing and digitizing the fingerprint of an individual for further processing.
- Traditionally, the **ink-on-paper technique** has been used to capture the fingerprint information from a person. It is a simple - yet powerful - acquisition and recording technique that has been in use since the late 19th century.

- However, the primary reason for the popularity of fingerprint recognition, particularly in non-forensic applications, is the **availability of mature, convenient, and low-cost sensors** that can rapidly acquire the fingerprint of an individual with minimum or no intervention from a human operator.
- These compact fingerprint sensors have also been embedded in many consumer devices such as laptops and mobile phones.
- Many sensing techniques that have been developed for acquiring fingerprints from a subject.

Fingerprint Sensing Techniques

- Digital images of the fingerprints can be acquired using **off-line or on-line methods**.

Off-line Techniques

- Off-line techniques generally do not produce the digital image directly from the fingertip.
- Rather, the fingerprint is first transferred to a substrate that is subsequently digitized.
- For example, an inked fingerprint image, the most common form of off-line capture, is acquired by first applying ink to the subject's fingertip and then rolling or pressing the finger on paper, thereby creating an impression of the fingerprint ridges on paper.
- The impression is then scanned and digitized using a flatbed document scanner.
- While the acquisition of inked fingerprints is still in practice, it is both **infeasible and socially unacceptable** in the context of non-forensic applications.
- **Latent fingerprint** development is another example of an off-line method. Latent fingerprints are lifted from the surface of objects that are touched or handled by a person. This is achieved through a variety of means ranging from simple photography to more complex dusting or chemical processing.

On-line Techniques

On-line techniques produce the **digital image** directly from a subject's fingertip via digital imaging technology that circumvents the need for obtaining an impression on a substrate. The resulting fingerprint image is referred to as a **live-scan fingerprint**.

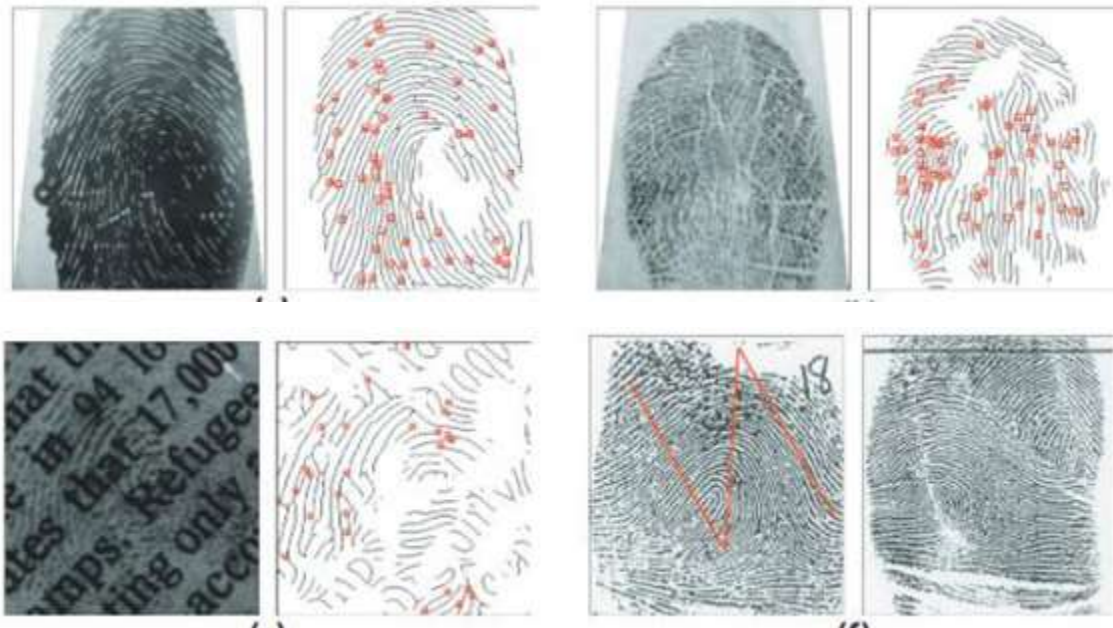
Some of the On-line techniques are

1. Optical Frustrated Total Internal Reflection
2. Capacitance based sensing
3. Ultrasound based sensing
4. Temperature based sensing
5. RF based sensing

Image Quality

Precise extraction of fingerprint features is crucially important for any automated as well as manual recognition system to run a match.

A good quality fingerprint image has distinguishable patterns and features that allow the extraction of features that are useful for subsequent matching of fingerprint pairs. Fingerprint image quality can be influenced by several factors like environmental, behavioral, quality of biometrics presented, system ability, etc. Some of the factors, affecting the fingerprint image quality and eventually the matching performance, can be addressed at user end, for example, subject might have wet fingers rendering biometric system unable to scan, so drying up fingers prior to sample collection will enhance quality of captured image. Environmental factors like operating the equipment in extreme environmental conditions; scratched or smudged scanning surface, etc., can also be addressed at operator or user end.



Poor Quality Images

Efforts for improving the quality of the image

1. **Live scan or electronic capture of fingerprint** is not only faster and efficient, but also produces clearer, sharper and high quality fingerprint images over ink-and-roll method.
2. **Controlling environmental conditions, user behavior and familiarizing users with acquisition process** can make a difference in sampled fingerprint image quality, for example, rubbing dirty fingertips with alcohol or drying up wet fingers will result in improved quality of sampled data. The scanning surface should also be cleaned and scratch free. Equipment should be placed away from unnecessary disturbances or extreme environmental conditions.
3. At technological end, biometric recognition system should also be able to put **image quality checks** in place and reject poorly scanned images. The recognition algorithm can be set to look for detail thresholds to accept or reject the sampled data and ask for re-sampling.

2. Feature Extraction

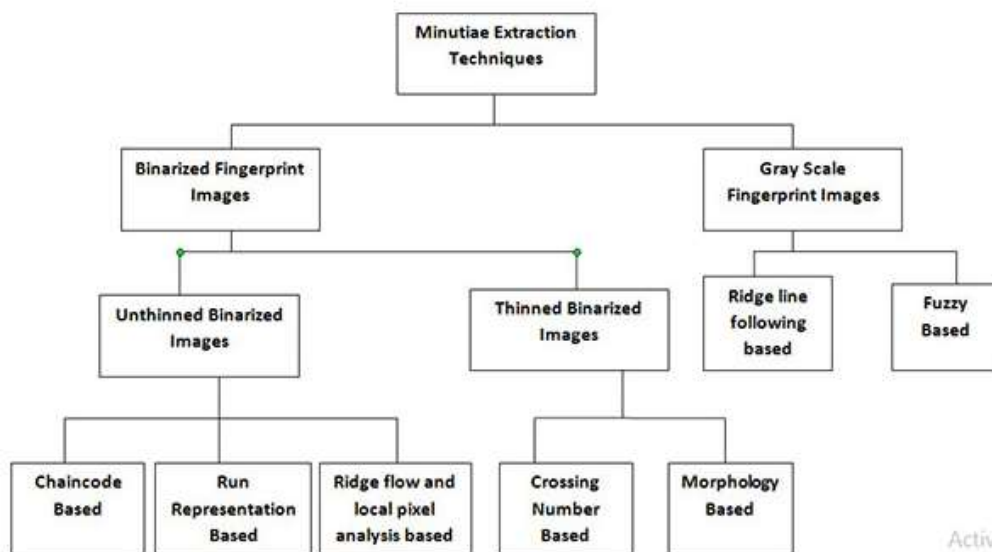
Fingerprint **feature extraction techniques** refer to the methods used to identify unique characteristics from a fingerprint image for recognition. These techniques can be broadly classified into three main categories:

1. Minutiae-Based Techniques (Most widely used) - Minutiae are small details in fingerprint ridges, such as ridge endings and bifurcations.
2. Pattern-Based (Global) Techniques- Uses the overall structure of the fingerprint rather than fine details.
3. Image-Based (Texture Analysis) Techniques- Extracts features directly from the gray-level image without binarization.
4. Hybrid Techniques- Combination of minutiae and texture features to improve robustness.

Minutiae Extraction Techniques

It is the most widely used technique of fingerprint representation and its configuration is highly distinctive. It is more accurate compared to other correlation based systems and the template size is smaller in minutiae based fingerprint representation. In this system, two fingerprints match if their minutiae points match. Minutiae based fingerprint technique is the backbone of most currently available fingerprint recognition products.

The minutiae extraction techniques are classified into



Gray scale image

A **grayscale fingerprint image** is an image where each pixel represents a shade of gray (0 to 255), instead of color. This is the most common format for fingerprint processing before binarization.



Binarized finger print images

Binarized fingerprint images are the result of converting a grayscale fingerprint image into a **binary format**, where each pixel is either black (ridge) or white (valley).



Image thinning (Skeletonization)

Image Thinning is an image processing operation used to reduce the width of objects (like fingerprint ridges) to a single-pixel thickness while preserving their essential structure and connectivity.

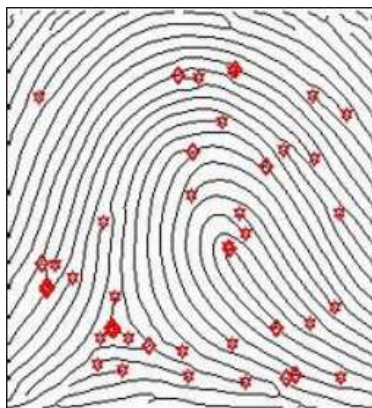


Minutiae Extraction

Minutiae extraction from a thinned image is the step where unique ridge characteristics like ridge endings and bifurcations are detected after the fingerprint ridges have been reduced to a single-pixel width.

A thinned fingerprint image has ridges of **one-pixel thickness**, which:

- Preserves ridge connectivity.
- Makes it easier to identify exact points where ridges end or split.
- Avoids false minutiae due to thick ridges.



Crossing number based Minutiae Recognition

Crossing-number (CN) is one of the simplest and most widely used methods for extracting minutiae (ridge endings and bifurcations) from a thinned fingerprint image.

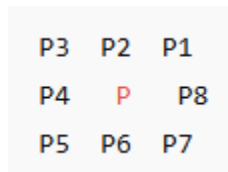
After we binarize and thin a fingerprint (so ridges are one-pixel wide), the crossing number method inspects each ridge pixel's 8-neighbourhood and counts how many times the pixel-to-neighbor value changes as you go once around the pixel.

- $CN = 1 \rightarrow$ ridge ending (termination)
- $CN = 3 \rightarrow$ bifurcation (one ridge splits into two)
- $CN = 2 \rightarrow$ ordinary ridge pixel (continuation)
- $CN = 4$ (or other values) $\rightarrow$ typically crossings or artefacts

The Crossing Number CN is:

$$CN = \frac{1}{2} \sum_{i=1}^8 |P_i - P_{i+1}| \quad \text{with } P_9 = P_1$$

Here each P_i is 1 for a ridge pixel and 0 for background.



P3	P2	P1
P4	P	P8
P5	P6	P7

Eg:

Example 1 — a ridge ending: only the right neighbor (P1) is 1, others 0.

Neighbors $P1..P8 = [1,0,0,0,0,0,0,0]$

Append P1 at end to form circular sequence: $[1,0,0,0,0,0,0,0,1]$

Compute absolute differences:

$$|P1-P2| = |1-0| = 1$$

$$|P2-P3| = |0-0| = 0$$

$$|P3-P4| = |0-0| = 0$$

$$|P4-P5| = |0-0| = 0$$

$$|P5-P6| = |0-0| = 0$$

$$|P6-P7| = |0-0| = 0$$

$$|P7-P8| = |0-0| = 0$$

$$|P8-P9| = |0-1| = 1$$

$$\text{Sum} = 1 + 0 + 0 + 0 + 0 + 0 + 0 + 1 = 2$$

$$\text{CN} = 0.5 \times 2 = 1 \rightarrow \text{ridge ending}$$

Example 2 — a bifurcation: P1, P3, P5 = 1 (three branches)

Neighbors = [1, 0, 1, 0, 1, 0, 0, 0]

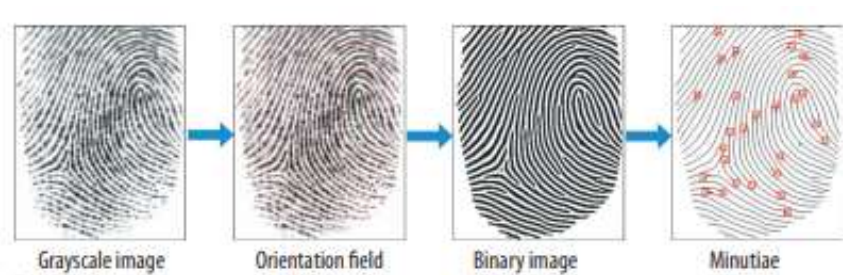
Circular sequence ... + [1]

Absolute diffs:

$$|1-0|=1, |0-1|=1, |1-0|=1, |0-1|=1, |1-0|=1, |0-0|=0, |0-0|=0, |0-1|=1$$

$$\text{Sum} = 1+1+1+1+1+0+0+1 = 6$$

$$\text{CN} = 0.5 \times 6 = 3 \rightarrow \text{bifurcation}$$



Flowchart of Minutiae feature extraction algorithm

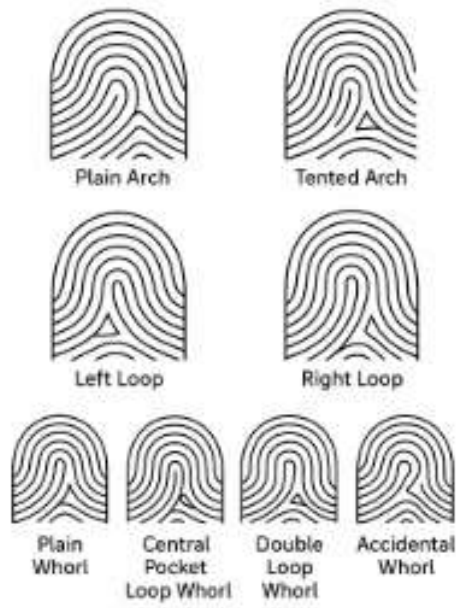
3. Finger print Classification

In the fingerprint recognition pipeline, *classification* is a stage that comes after feature extraction and before matching/identification. Its role is to categorize the fingerprint into one of the standard fingerprint classes so that the search space is reduced during identification. i.e, Assign the fingerprint into a class based on ridge flow pattern.

Fingerprint pattern classes

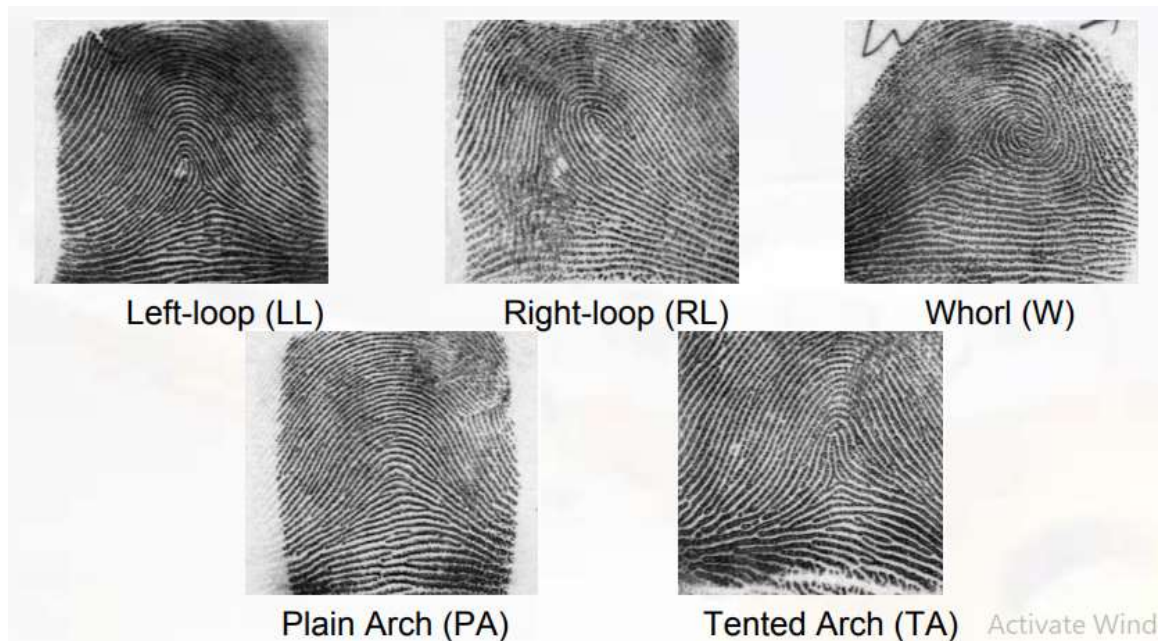
The fingerprint can be represented as 8 detailed sub-classes

Fingerprint Pattern Classes



Henry (1900) made an extensive study of occurrence of fingerprints and indexed them into 8 major classes. The 8 classes are shown above.

The four different whorl classes can be combined into one class: Whorl (W). It can be represented as



The Henry system with five classes are shown above. The five classes can be reduced to four by combining the PA and TA classes to form the Arch (A) class. The natural frequencies of W, L, R and A (A + T) are 27.9%, 33.8%, 31.7% and 6.6%.

The five main classes differ in terms of the global flow patterns of the ridge curves.

- They also differ in terms of the number and locations of singular points in the fingerprint image. For example,

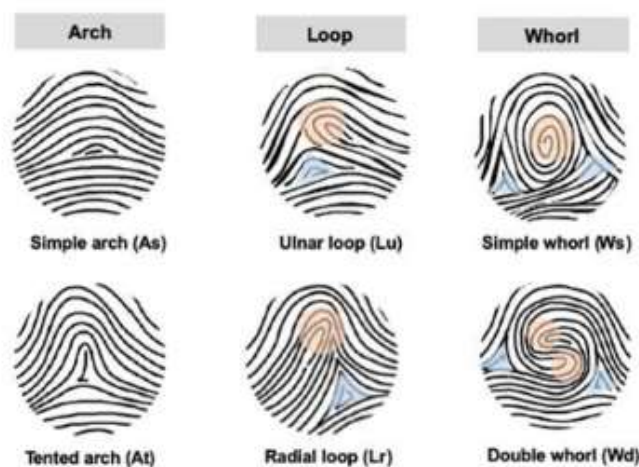
LL – exactly one core and one delta; the core is to the left of the delta,

RL – exactly one core and one delta; the core is to the right of the delta,

W – Two cores and two deltas,

PA – no singular points, and

TA – one core and one delta; the delta is approximately directly below the core.



Classes showing loops and deltas

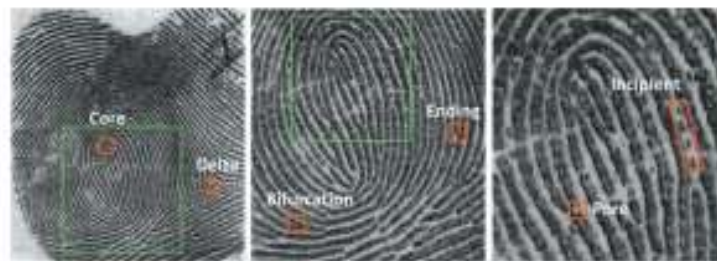
Fingerprint Classification Approaches

Fingerprint classification can be done using different approaches, depending on whether it is a traditional (rule-based) or modern (machine learning/deep learning) system.

1) Rule based approaches

These are the traditional methods where classification is done using explicit rules derived from fingerprint features such as singular points, ridge orientation, and ridge counts.

Eg., detecting singular points(Core and Delta)



Pointcare Index method

In fingerprint recognition, singular points (core and delta) are critical for:

- Fingerprint classification (arch, loop, whorl, etc.)
- Alignment of fingerprints before minutiae matching
- Reliable feature extraction

The Poincaré Index Method is a mathematical approach used to locate these singular points by analyzing the **orientation field** of the fingerprint ridges.

Steps:

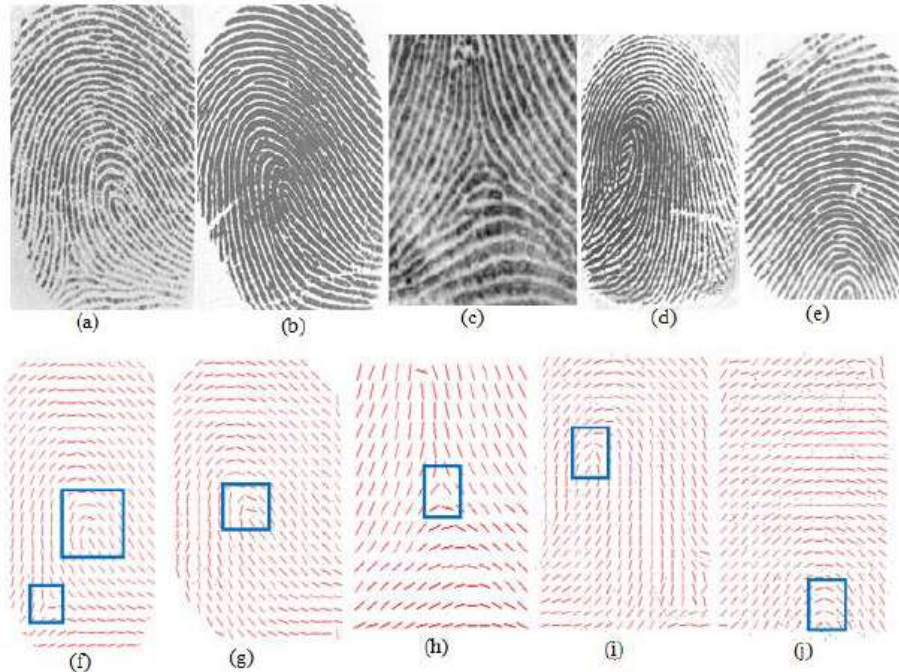
- Computation of orientation field using gradient method

A fingerprint image is divided into non-overlapping blocks (commonly 8×8 or 16×16 pixels). For each block, the ridge orientation $\theta(i,j)$ is computed using gradients:

$$\theta(i,j) = \frac{1}{2} \tan^{-1} \left(\frac{2 \sum G_x G_y}{\sum (G_x^2 - G_y^2)} \right)$$

where G_x, G_y are image gradients along x and y.

This will give the orientation field.



- Calculation of point care index using orientation differences

The **Poincaré Index (PI)** measures how much the orientation changes around a **closed loop** (usually 8 blocks around a central block).

$$PI = \frac{1}{2\pi} \sum_{k=1}^n \Delta\theta_k$$

where:

- $n = 8$ (neighbors in a loop)
- $\Delta\theta_k = \theta_{k+1} - \theta_k$ (difference in orientation between adjacent neighbors)
- Adjust $\Delta\theta_k$ to be within $(-\pi/2, +\pi/2)$ to avoid jumps.

- Interpretation of values

If the **PI** = $+\frac{1}{2}$ (**i.e., +180°**) → Core point

- Found in loop and whorl patterns.
- Represents a ridge "turning" center.

If the **PI** = $-\frac{1}{2}$ (i.e., -180°) $\rightarrow$ Delta point

- Found in loop and whorl patterns.
- Represents a ridge diverging point.

If the **PI** = **0** $\rightarrow$ No singularity (plain region).

If the **PI** = ± 1 $\rightarrow$ Double core or rare singularities.

Example:

Consider a **3×3 block region** where we want to check if the **center block** is a singular point.

θ_1	θ_2	θ_3
θ_8	C	θ_4
θ_7	θ_6	θ_5

Where

- C = central block we are testing
- $\theta_1 \dots \theta_8$ = orientation angles of surrounding 8 blocks (arranged counterclockwise around C).

Collection of orientation values

Assume the ridge orientations (in degrees) around the central block are:

$$\theta_1 = 10^\circ, \theta_2 = 40^\circ, \theta_3 = 80^\circ, \theta_4 = 120^\circ, \theta_5 = 160^\circ, \theta_6 = -160^\circ, \theta_7 = -120^\circ, \theta_8 = -80^\circ$$

We compute $\Delta\theta = \theta_{k+1} - \theta_k$, making sure each difference stays within $(-90^\circ, +90^\circ)$.

- $\Delta\theta_1 = 40 - 10 = +30^\circ$
- $\Delta\theta_2 = 80 - 40 = +40^\circ$
- $\Delta\theta_3 = 120 - 80 = +40^\circ$
- $\Delta\theta_4 = 160 - 120 = +40^\circ$
- $\Delta\theta_5 = (-160) - 160 = -320 \rightarrow \text{adjust} \rightarrow +40^\circ$
- $\Delta\theta_6 = (-120) - (-160) = +40^\circ$
- $\Delta\theta_7 = (-80) - (-120) = +40^\circ$
- $\Delta\theta_8 = 10 - (-80) = +90^\circ$

$$\text{Total change} = 30 + 40 + 40 + 40 + 40 + 40 + 40 + 90 = 360^\circ$$

Computation of pointcare index

$$PI = \frac{1}{2\pi} \sum \Delta\theta = \frac{360^\circ}{360^\circ} = +1$$

As PI is +1, core point is detected at the centre.

2) Statistical Methods

In these approaches, the fingerprint image is represented as numerical data (feature vectors), and statistical decision-making is applied to classify the fingerprint into a class (arch, loop, whorl, etc.).

Instead of hard rules (like in rule-based), they use mathematical models, probability, and similarity measures.

Eg. Ridge orientation analysis, Ridge frequency analysis and ridge count.

3) Machine learning approaches

Unlike rule-based (explicit rules) or statistical (feature distributions), machine learning (ML) methods learn classification rules from data.

They take features (like orientation, ridge frequency, singular points, or even raw images) and train a model to predict the class (arch, loop, whorl, etc.).

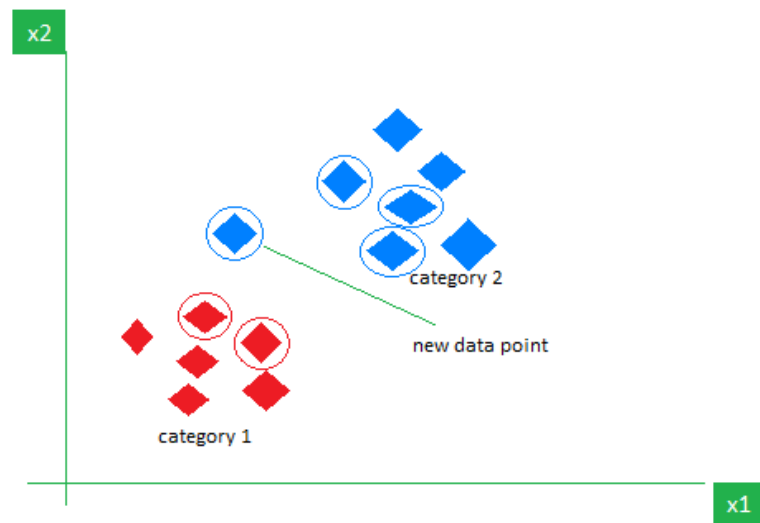
Eg. K-nearest neighbor and support vector machine.

K-nearest neighbor algorithm

Knn is a non-parametric supervised learning technique in which we try to classify the data point to a given category with the help of training set. In simple words, it captures information of all training cases and classifies new cases based on a similarity.

It is also called an **instance-based learning** algorithm because predictions are made based on stored instances.

Predictions are made for a new instance (x) by searching through the entire training set for the K most similar cases (neighbors) and summarizing the output variable for those K cases. In classification this is the mode (or most common) class value.



Here are the basic steps involved in the KNN algorithm:

1. Choose the Number of Neighbors (k)

- Select the value of k, which represents the number of nearest neighbors to consider for making predictions.

2. Prepare the Data

- Split the dataset into training and test sets.
- Normalize or standardize the data if necessary to ensure that all features contribute equally to the distance calculations.

3. Calculate Distances

- For each instance in the test set, compute the distance to all instances in the training set. Common distance metrics include:
 - Euclidean distance

Euclidean distance is defined as the straight-line distance between two points in a plane or space. You can think of it like the shortest path you would walk if you were to go directly from one point to another.

$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{ij})^2}$$

- Manhattan distance
- Minkowski distance

4. Identify Nearest Neighbors

- Sort the calculated distances and identify the k closest training instances to the test instance.

5. Make Predictions

- For classification:
 - Assign the class label that is most common among the k nearest neighbors.
- For regression:
 - Calculate the average (or weighted average) of the target values of the k nearest neighbors.

6. Evaluate the Model

- Assess the model's performance using metrics such as accuracy, precision, recall, or mean squared error, depending on whether the task is classification or regression.

These steps outline the fundamental process of the KNN algorithm, which is widely used due to its simplicity and effectiveness in various applications.

Fingerprint Matching

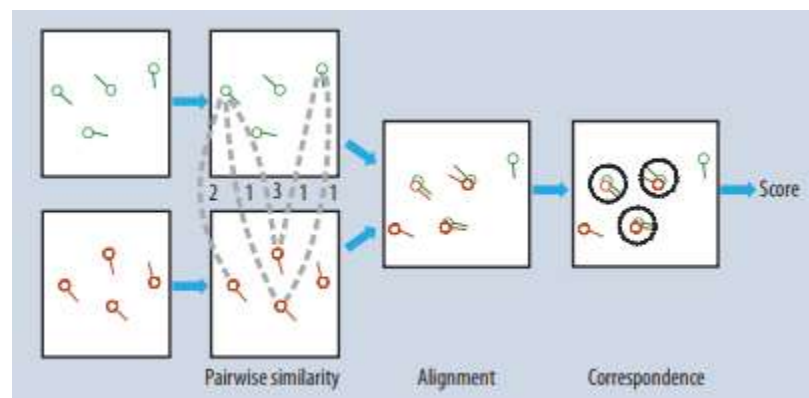
A fingerprint matching module computes a match score between two fingerprints, which should be high for fingerprints from the same finger and low for those from different fingers. Fingerprint matching is a difficult pattern-recognition problem due to large intraclass variations (variations in fingerprint images of the same finger) and large interclass similarity (similarity between fingerprint images from different fingers). Intraclass variations are caused by finger pressure and placement—rotation, translation, and contact area—with respect to the sensor and condition of the finger

such as skin dryness and cuts. Meanwhile, interclass similarity can be large because there are only three types of major fingerprint patterns (arch, loop, and whorl).

Most fingerprint-matching algorithms adopt one of four approaches: image correlation, phase matching, skeleton matching, and minutiae matching. Minutiae-based representation is commonly used, primarily because

- Forensic examiners have successfully relied on minutiae to match fingerprints for more than a century,
- Minutiae-based representation is storage efficient, and
- Expert testimony about suspect identity based on mated minutiae is admissible in courts of law.

The current trend in minutiae matching is to use local minutiae structures to quickly find a coarse alignment between two fingerprints and then consolidate the local matching results at a global level. This kind of matching algorithm typically consists of four steps, the figure shows.



Minutiae Matching Algorithm

First, the algorithm computes pairwise similarity between minutiae of two fingerprints by comparing minutiae descriptors that are invariant to rotation and translation. Next, it aligns two fingerprints according to the most similar minutiae pair. The algorithm then establishes minutiae correspondence—minutiae that are close enough both in location and direction are deemed to be corresponding (mated) minutiae. Finally, the algorithm computes a similarity score to reflect the degree of match between two fingerprints based on factors such as the number of matching minutiae, the percentage of matching minutiae in the overlapping area of two fingerprints, and the consistency of ridge count between matching minutiae

Ridge feature based fingerprint matching using Gabor filters

Ridge Feature-Based Fingerprint Matching Using Gabor Filters is a biometric technique used to match fingerprint patterns by analyzing ridge structures (such as ridge orientation and frequency) with the help of Gabor filters, which are powerful tools for texture analysis and feature extraction.

Fingerprint ridge features are the structural and textural patterns formed by the **ridges** (raised lines) and **valleys** (spaces between ridges) in a fingerprint. These features are essential in fingerprint recognition, especially in **ridge-based matching systems**, where the entire ridge flow and texture are used instead of just minutiae points.

Types of finger print ridge features are:

1. Ridge Orientation
2. Ridge Frequency
3. Ridge strength/Ridge energy
4. Ridge Continuity
5. Ridge curvature
6. Texture patterns

Application of Gabor filter

Gabor filters have been a cornerstone in the field of image processing and machine learning for decades. Named after Dennis Gabor, who introduced them in the 1940s, these filters have proven to be incredibly versatile and powerful tools for analyzing and manipulating visual data.

These are a type of linear filter used for edge detection and texture analysis. They are defined by a sinusoidal function modulated by a Gaussian distribution. The mathematical formulation of a Gabor filter can be represented as a complex-valued function:

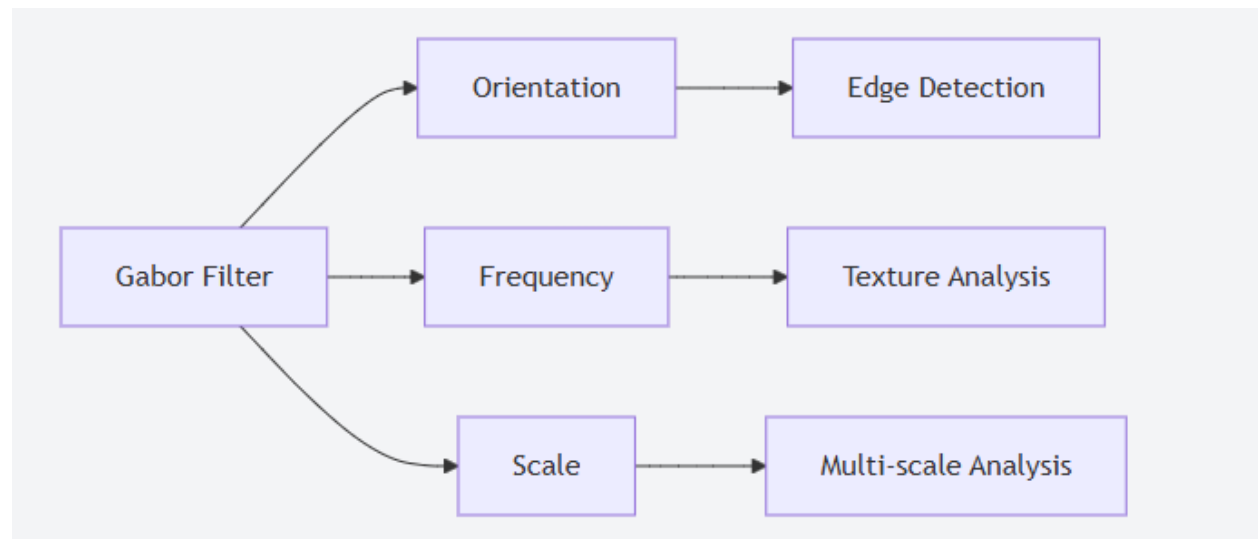
$$G(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \exp\left(i\left(2\pi\frac{x'}{\lambda} + \psi\right)\right)$$

where $x' = x \cos \theta + y \sin \theta$ and $y' = -x \sin \theta + y \cos \theta$. The parameters λ , θ , ψ , σ , and γ control the frequency, orientation, phase, scale, and aspect ratio of the filter, respectively.

Properties of Gabor Filters

Gabor filters have several important properties that make them useful for image processing tasks:

- **Orientation:** Gabor filters can be designed to be sensitive to specific orientations, making them useful for detecting edges and textures at particular angles.
- **Frequency:** Gabor filters can be tuned to specific frequencies, allowing them to capture the intrinsic properties of textures and patterns in images.
- **Scale:** Gabor filters can be scaled to capture features at different sizes, making them useful for multi-scale image analysis.



Finger print enhancement and matching algorithms

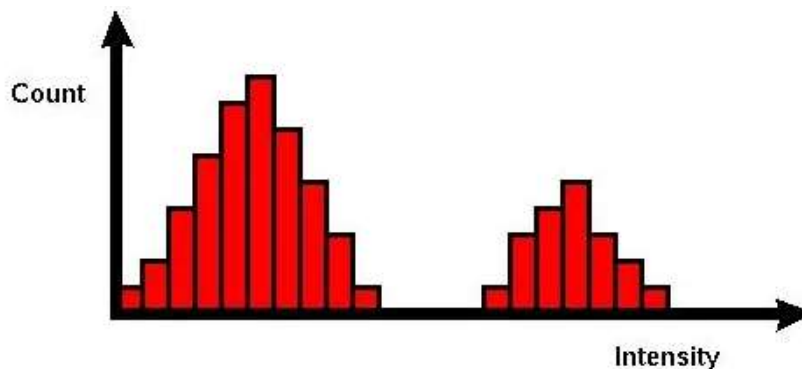
Enhancement algorithms

Fingerprint image enhancement is an essential preprocessing step in fingerprint recognition applications and is a technique performed to make the image clearer than the original image. Fingerprint recognition system using enhancement algorithm methods that needed to increase the contrasting between ridges and valleys and for connection the false broken points of ridges due to insufficient mount of black color (ridgelines).

Enhancement algorithm is an important step in any applications execution based authentication system and used to correct noisy in fingerprint surface. Ridgelines are present by black or dark gray color and valleys gap narrow is present by white color. The noises, errors, and broken ridges are arise with a poor skin condition, varying finger pressure while acquisition or enrollment, sensor noise and dry, wet or oily fingers. Enhancement algorithm is a preprocessing algorithm work with original image to created clearer ridges and valleys before extract features of minutiae.

Histogram equalization

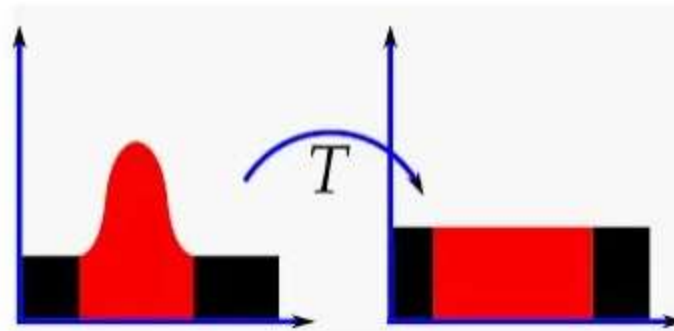
A histogram of an image is the graphical interpretation of the image's pixel intensity values. It can be interpreted as the data structure that stores the frequencies of all the pixel intensity levels in the image.



As we can see in the image above, the X-axis represents the pixel intensity levels of the image. The intensity level usually ranges from 0 to 255.

Histogram Equalization is an image processing technique that adjusts the contrast of an image by using its histogram. To enhance the image's contrast, it spreads out the most frequent pixel intensity values or stretches out the intensity range of the image.

By accomplishing this, histogram equalization allows the image's areas with lower contrast to gain a higher contrast.



Filtering Methods

Gabor filters

Widely used to enhance ridge orientation and frequency. They are orientation- and frequency-sensitive filters that highlight ridges while suppressing noise.

Fourier Transform based Filtering

Fingerprints have a periodic ridge–valley pattern. The ridges are like sinusoidal waves that repeat at a certain frequency and orientation.

- In the spatial domain (image view), ridges look like alternating dark and light lines.
- In the frequency domain (Fourier transform view), these repeating ridge patterns appear as distinct frequency components.

By applying the **Fourier Transform (FT)**, we can move from the **spatial domain** to the **frequency domain**, where filtering becomes easier.

The 2D Fourier Transform of an image $f(x, y)$ is:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Where:

- $(x, y) \rightarrow$ spatial coordinates (pixels).
- $(u, v) \rightarrow$ frequency domain coordinates.
- $F(u, v) \rightarrow$ frequency representation of the image.

Wavelets for fingerprint

Fingerprints contain fine, repeating ridge–valley patterns that may vary in thickness, curvature, and orientation across the image. Unlike Fourier transform (which gives only frequency information but loses spatial details), the wavelet transform provides both **spatial (location) and frequency (scale) information**. This makes wavelets ideal for enhancing localized ridge details while suppressing noise.

Image Binarization

The black and white image (called binary image) are transforming into two parts (object and background). The object is often shown as black and background is often shown as white.

Binarization technique will be analysis on each pixel of image after calculating the average pixels. To examine the average color the pixel will be compared with a threshold value. For eg. If the threshold calculated is 126. The pixel is black (or 1) if it is less than 126 and the pixel is white (or 0) if it is bigger than or equal 126. The methodology to divide these two parts is according to analysis based on if...then condition on threshold while the wide range of pixels is 255. If the color of each pixel are brightness than the average will determine the pixel as white and beside this will determine the pixel image to black.

Image Thinning

Thinning (skeletonization) is a morphological operation that reduces binary objects (ridges) to a one-pixel wide skeleton while preserving their structure and connectivity. In fingerprints, thinning makes ridges exactly one pixel thick, which is crucial for detecting minutiae points (ridge endings & bifurcations).

Most popular algorithm used for thinning is **Zhang–Suen Thinning Algorithm**.

The **Zhang–Suen Thinning Algorithm** is a well-known **skeletonization** algorithm used in **image processing** to reduce binary images (like fingerprints, text, or shapes) to their **thinned skeletons** while preserving connectivity and general structure.

Steps of Zhang–Suen Thinning Algorithm

The algorithm works iteratively in two sub-steps until no further pixel changes occur.

Input:

A **binary image**

- Foreground (object) pixels = 1 (black)
- Background pixels = 0 (white)

Conditions:

- For a pixel **P1** (candidate for deletion), define its **8-neighbors** in clockwise order:

```
P9 P2 P3
P8 P1 P4
P7 P6 P5
```

- **N(P1)**: Number of nonzero neighbors (black pixels) around P1.
- **S(P1)**: Number of $0 \rightarrow 1$ transitions in the ordered sequence (P2, P3, ..., P9, P2).

Iteration Steps:

Step 1:

A pixel **P1** is deleted if all conditions are satisfied:

1. $2 \leq N(P1) \leq 6$
2. $S(P1) = 1$
3. $P2 \times P4 \times P6 = 0$
4. $P4 \times P6 \times P8 = 0$

Step 2:

A pixel **P1** is deleted if:

1. $2 \leq N(P1) \leq 6$
2. $S(P1) = 1$
3. $P2 \times P4 \times P8 = 0$
4. $P2 \times P6 \times P8 = 0$

Repeat step 1 and 2 until no pixels can be removed.

Matching algorithms

Matching fingerprint are used minutiae on the fingerprint to compare with each other. The two fundamental principles immutability (ridge patterns never change during the lift time) and uniqueness (distinct ridge patterns on different fingers of the same individual) are used in identification of individual's fingerprint.

No match is perfect in both verification and identification. If match is found or not based on a threshold value by using real minutiae points to compare each other. It consists of finding the best alignment between the compared fingerprints before processing with the comparison step to maximize the score that quantified the quality of the matching. The fingerprint biometric has its own strength and limitations.

The spurious minutiae are the problem for accuracy of matching. Good spurious minutiae removal is the common key point for good matching minutiae. The most of existing fingerprint matching approaches can coarsely classified into two families based on different features: **minutiae based algorithms and global feature-based algorithms.**

Minutiae-based algorithm is to maximize the number of matching minutiae pairs between two fingerprints and the local minutia topologic structures are widely used in minutia matching methods since local areas suffer less from non-linear distortion. The most popular global feature-based are remains to fixed-length feature was extracted from the region around the reference point after filtering by a bank of Gabor filters that response to different ridge orientation. The most commonly used techniques for fingerprint matching are divided into three approaches as the following:

1) **Minutiae-based:** Minutiae is a majority for fingerprint identification technique. It is the identification of minutiae points along with their relative position on finger. Minutiae refers to the points where the ridgelines terminate or fork are call minutiae whereas according to Galton, each ridge is characterized by numerous minute peculiarities and there are two fingerprints match if their minutiae are matched.

2) **Correlation-based:** This technique using match two fingerprints are aligned and the correlation is computed for corresponding pixels, however, as the displacement and rotation are unknown it is necessary to apply the correlation for all possible alignments. It is based on abundant gray scale information but it can be bad with quality data.

3) **Pattern matching or Ridge feature based:** This technique are based on series of ridges as opposed to discrete points which forms the basis of ridge feature based. It compares the basic fingerprint patterns between claimant and a store fingerprint templates. A matching using ridges feature in form of finger code consists of computing the difference of two fingers code vectors (query and reference).

Among three matching algorithms approach, the minutiae-based algorithms is the most recommended and very sensitive to the accuracy of the minutia detection and orientation computation. Minutia feature is only using for matching.